AMERY ICE SHELF BOREHOLE MICROCAT AND ADCP DATA, 2012/2013 SEASON - DATA PROCESSING AND QUALITY

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INTRODUCTION

This report summarises the latest AMISOR oceanographic data, season 2012/2013, collected from oceanographic equipment deployed through the Amery Ice Shelf boreholes. Time series data of temperature, salinity and pressure were downloaded from two strings of Seabird SBE37IM microcats at AMISOR boreholes AM05 (10.5 months of data) and AM06 (10.5 months of data). No data were obtained from the ADCP (RDI 300 kHz broadband Workhorse) on the AM06 string, due to power supply problems. Data processing and data quality details are included in the report.

Table 1: Station details for AMISOR season 2012/2013 borehole microcats. Note that borehole absolute locations change in time, as the ice shelf is in motion. For the microcats, positive clock errors indicate microcat clocks were fast. Instrument pressures/depths are mean values from the pressure sensors over the data dates (including all pressure steps), with the exception of microcats 7327 and 7325, where the pressure/depth is a point value from soon after initial deployment.

Borehole microcats

borehole	position at given date		microcat	instrument pos.		data dates		rec.	clock
	latitude	longitude		depth (m)	pressure (dbar)	start	end	interval (min.)	error (sec.)
AM05	70°13.369'S	69°41.392'E	7327	573.5	580.2	24-Feb-2012	11-Jan-2013	30	-11
AM05	(on 11-Jan-2013)		7326	693.2	701.5	24-Feb-2012	11-Jan-2013	30	+6
AM05	"	"	7322	828.7	838.9	24-Feb-2012	11-Jan-2013	30	-3
AM06	70°14.216'S	71°22.644'E	7325	545.0	551.3	24-Feb-2012	11-Jan-2013	30	+6
AM06	(on 11-Jan-2013)		7324	671.5	679.5	24-Feb-2012	11-Jan-2013	30	+17
AM06	"	"	7323	780.5	790.0	24-Feb-2012	11-Jan-2013	30	+4

Table 2: Microcat pressure sensor ratings

borehole	microcat	sensor rating (dbar)	borehole	microcat	sensor rating (dbar)
AM01	1969	1000	AM04	1972	1000
AM01	1970	1000	AM04	1973	1000
AM01	1971	1000	AM04	1974	3500
AM02	1623	1034	AM05	7327	1000
AM02	1624	1034	AM05	7326	1000
AM02	1174	2068	AM05	7322	2000
AM03	4055	1000	AM06	7325	1000
AM03	4054	1000	AM06	7324	1000
AM03	3883	2000	AM06	7323	2000

MICROCAT DATA

The microcats were downloaded by Adam Christensen (brief field report available). The microcat moorings have already been described in previous field and data reports. This was the third download

of data from the AM05 and AM06 instruments (deployed in December 2009 and January 2010 respectively). All latest microcat data downloaded during the 2012/2013 season (Table 1) were assigned the same decimal time scheme consistent with previous data i.e. decimal days as counted from midnight on December 31st 2000 (so, e.g. midday on January 1st 2001 = 0.5 decimal time; midday on January 1st 2002 = 365.5).

The microcats were downloaded using the SeaBird terminal program Seaterm, and instrument clock errors (Table 1) were obtained from the "debug" instrument communication files. No clock drift corrections were applied as clock errors were small. Microcat clocks were all reset to GPS time following data download and prior to reinitialisation of logging.

Manufacturer supplied calibrations (September 2009 for AM05 and AM06) were applied internally by the microcats, and calibrated data were output. The data were then processed, as follows:

* The files were padded at the start and end, data gaps were checked for and filled, and decimal times were calculated.

* Mean pressure/depth values were calculated for the year's data for each microcat (Table 1), with all pressure steps included in the means.

* Pressure data corrections were required for the shallowest microcat at AM05 and AM06 (#7327 and #7325 respectively), and conductivity/salinity data were recalculated with the corrected pressures (more details given below).

Microcat data quality notes

A whole range of data features are present in the current season's download, from interesting real data features, to suspect but unexplained data, to obviously bad data.

The following data spikes are present in the data (not removed from the files):

borehole	microcat	spike	times
AM05	7326	temp/sal spike	0100 on 01/03/2012
AM06	7325	cond/sal fouling spike	1700 on 12/06/2012
AM06	7324	cond/sal fouling spike	0200 on 12/07/2012
AM06	7324	cond/sal fouling spike	0300 on 20/07/2012
AM06	7324	cond/sal fouling spike	0330 to 0400 on 27/07/2012
AM06	7324	cond/sal fouling spike	0530 on 27/07/2012
AM06	7324	cond/sal fouling spike	0500 on 06/10/2012
AM06	7323	cond/sal fouling spike	2330 on 06/08/2012
AM06	7323	cond/sal fouling spike	2000 on 10/09/2012

Microcat data are plotted in Figures 1 to 3. From the pressure sensor data (Figure 3), mooring depth changes are evident at AM05, as indicated by pressure steps. The larger step is most likely due to increased buoyancy from ice accretion on the mooring line. Smaller steps or trends may be due to ice shelf deformation. Strong current events, causing temporary "push-up" of the instruments and as indicated by pressure spikes (Figure 3), are larger at AM06 than at AM06.

For AM05, there is a strong correlation between microcats 7326 and 7322 for pressure drift and pressure step events (Figure 3), indicating these features are real. For microcat 7327, all pressure data are bad, and a single value from soon after deployment has been used to recalculate all conductivity and salinity data. Use of this single pressure value obviously misses the pressure events seen in the other 2 instruments (Figure 3), and as a result the pressure for cat7327 is almost certainly in error by ~50 dbar by the end of the time series (corresponding to the cumulative drift/shifts seen in the other 2 instruments since initial deployment in 2009/2010). This is equivalent to a salinity error of

~0.03 (PSS78). Note also that unexpected salinity and conductivity trends are evident in the recalculated cat7327 data, possibly real data measured by an instrument frozen in a cavity in the ice.

For AM06, all pressure data are bad for microcat 7325, and a single value from soon after deployment has been used to recalculate all conductivity and salinity data. Use of this single pressure value obviously misses any pressure events seen previously in the other 2 instruments, however no significant pressure steps are evident at AM06 (as seen at AM05). As a result the pressure for cat7325 is in error by a small amount only, ~0.5 dbar by the end of the time series (corresponding to the cumulative drift/shifts seen in the other 2 instruments since initial deployment in 2009/2010). Any equivalent salinity error is negligible. From the evidence of the bad microcat 7325 pressure data for the 2012/2013 data, in hindsight the microcat 7325 pressure data for 2011/2012 are also bad, and the data files using the single pressure value (i.e. to recalculate conductivity and salinity) are the files that should be used (files cat7325_2.dat and cat7325_2.mat from the 2011/12 data set).

For AM06, pressure data for the middle microcat 7324 drifts by ~0.5 dbar over the year, with values decreasing. As there is no corresponding drift for the deep microcat, this is assumed to be a calibration drift.

In addition to any errors described above, from conductivity and temperature sensor calibration drift estimates as specified by the manufacturer (0.0003 (PSS78) **per month** for salinity, and 0.0005^{o}C **per year** for temperature), we might expect the following calibration drifts at the time of the latest 2012/2013 data downloads:

AM05/AM06 (in water for 3 years) total S drift = 0.011 (PSS78) total T drift = 0.0015^{o}C

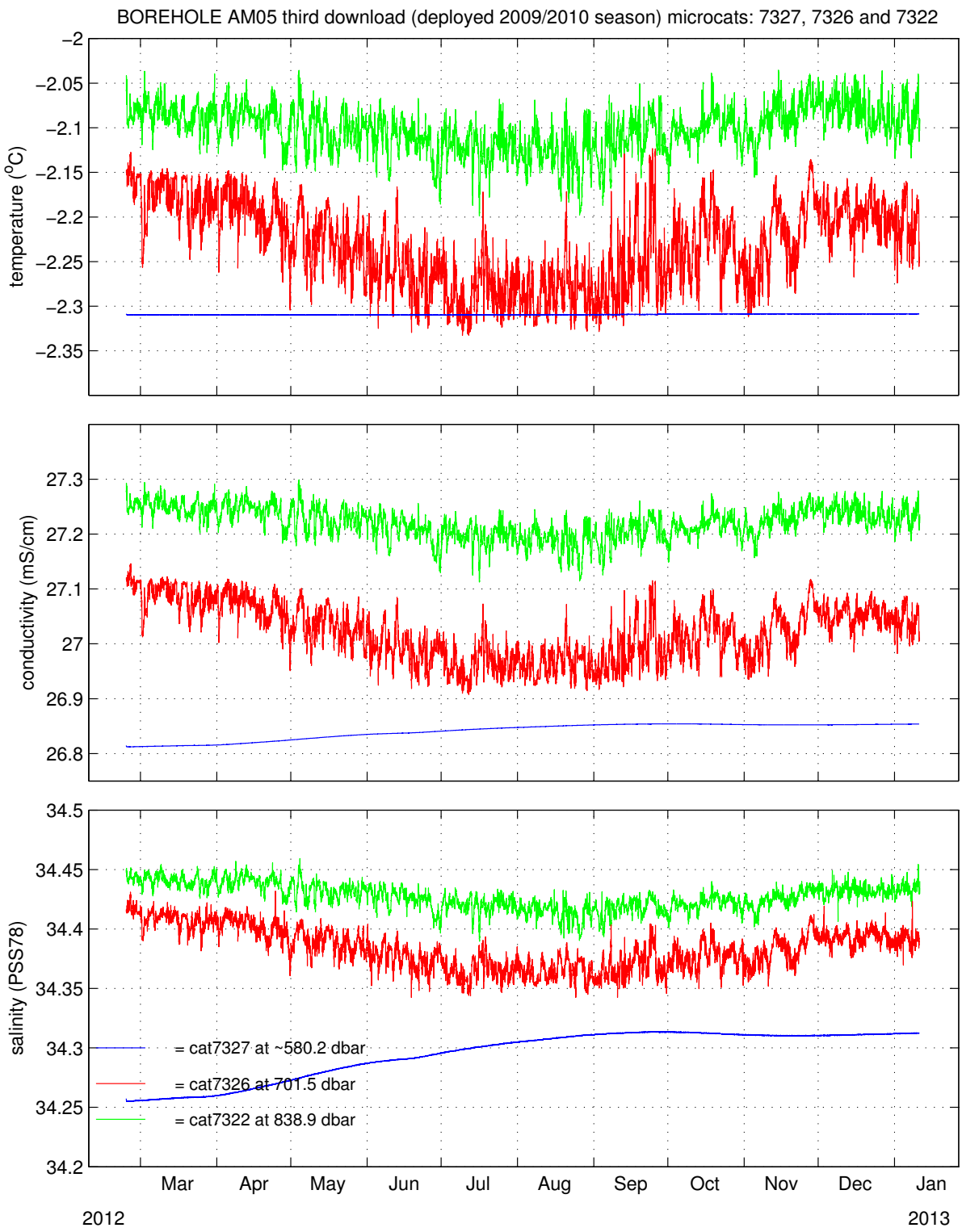


Figure 1: Borehole AM05 microcat data downloaded in 2012/2013 season (note: single large temperature/salinity spike for cat7326 not shown).

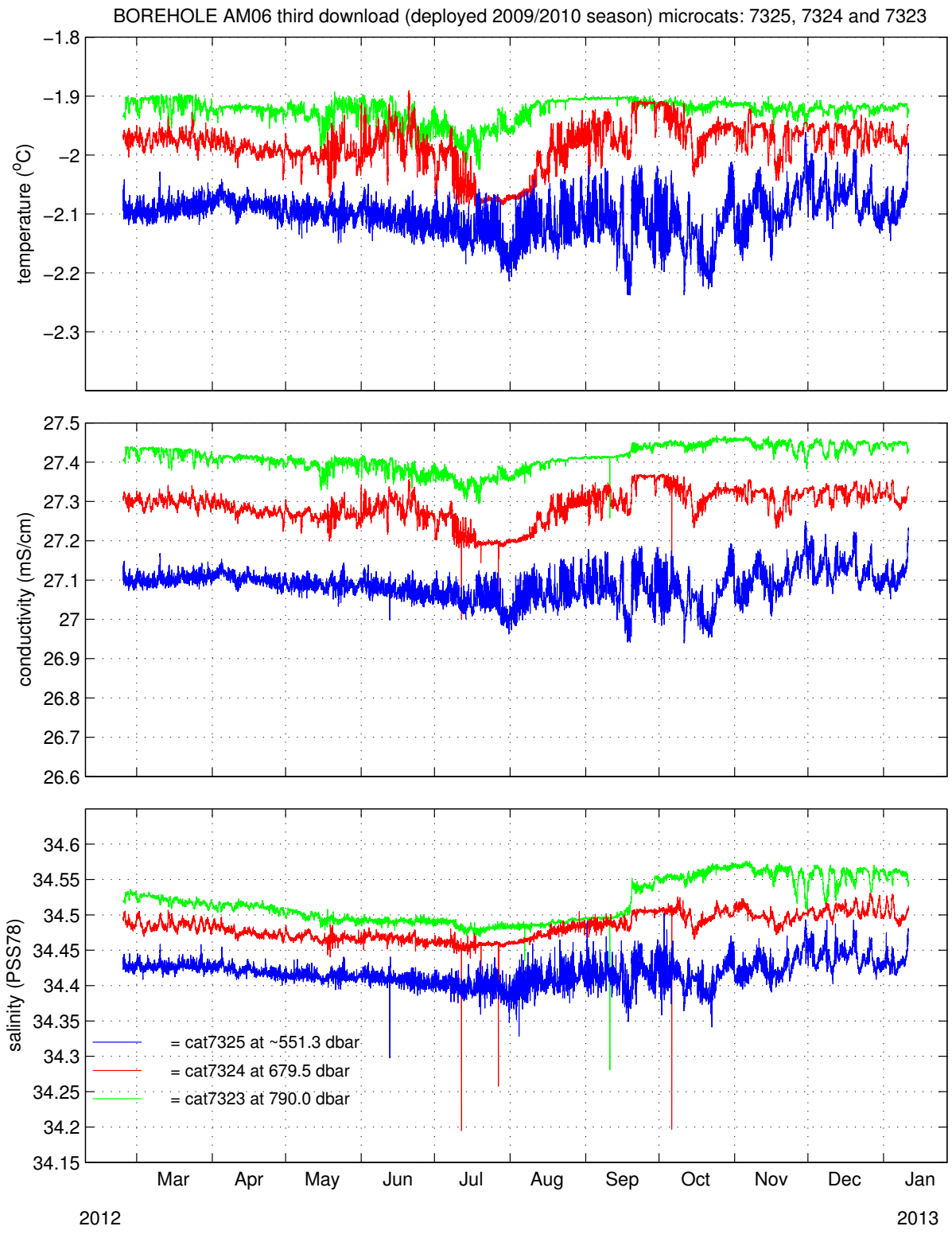


Figure 2: **Borehole AM06 microcat data downloaded in 2012/2013 season.**

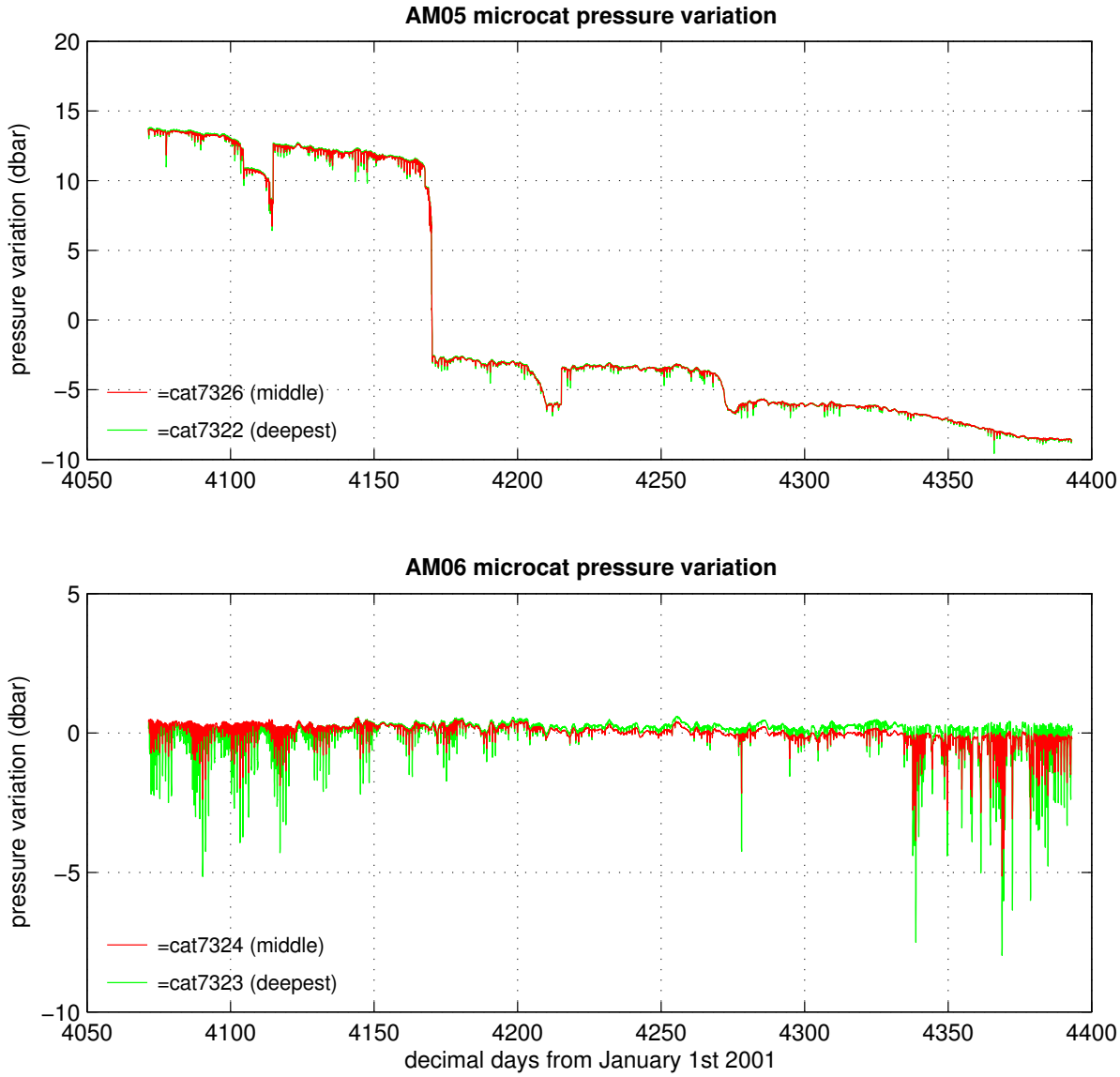


Figure 3: Pressure sensor data from boreholes AM05 and AM06 microcats downloaded in 2012/2013 season. Plotted pressures are variations about the mean value. Note: pressures for the shallowest microcat at AM05 and AM06 are not shown (bad data).

DATA FILES

New microcat data from boreholes AM05 and AM06 are contained in text and matlab format files, as follows:

1. Microcat data

Note: at the time of writing, the unified microcat files (i.e. all data from initial deployment time) have not been updated. New data files only contain the latest downloaded data (2012/13).

AM05

Data from the third data download (i.e. 24/02/2012-11/01/2013) are contained in the files:
cat7327_3.dat, cat7326_3.dat, cat7322_2.dat (text files), and
cat7327_3.mat, cat7326_3.mat, cat7322_3.mat (matlab files).
README cat am05_3 details the microcat file formats.
Data in the cat7327_3 files use a single pressure value (from soon after initial deployment) for all salinity and conductivity calculations

AM06

Data from the third data download (i.e. 24/02/2012-11/01/2013) are contained in the files:
cat7325_3.dat, cat7324_3.dat, cat7323_3.dat (text files), and
cat7325_3.mat, cat7324_3.mat, cat7323_3.mat (matlab files).
README cat am06_3 details the microcat file formats.
Data in the cat7325_3 files use a single pressure value (from soon after initial deployment) for all salinity and conductivity calculations.

2. Station and summary info files

The files station.dat and borehole_data_summary.dat are updated to include the latest season's data.